PDR - Design Specification

Problem

Disabled adults may have difficulties with energy consumption and fatigue, and, as such, would have to choose carefully what their energy would be spent on, resulting in many everyday activities or chores being postponed in favour of more urgent or important tasks. While, for a non-disabled adult, taking time out of their day to take their trash out or wash their dishes would be a quick and easy endeavour, for disabled adults, these constant chores take more energy out of those with already limited energy.
One of the more consistent and challenging chores is feeding oneself - in addition to a lack of appetite due to chronic pain making eating difficult, the task of making sure one's food is nutritious as well as delicious can be arduous. Combined with the physical task of actually making the food, and it is clear why having consistent, good quality, home cooked meals often requires a personal carer.

However, personal carers also have their own lives to live, and both them and their charges should have more time to invest how they see fit. Being a carer is stressful, being a ward is stressful, and both parties deserve any breaks in stress that can be given.

Answer

The Personal Delivery Robot (PDR) is intended for use in a single house only, and aims to remove the work of planning, organising, and delivering healthy meals consistently to multiple people in the house.
By outsourcing these energy-taxing tasks, both carers and wards can spend more time on other activities. While the carer is still intended to cook meals, the PDR will handle delivering food (and other objects) to various locations throughout the living space.
Overall, the PDR should cost less money than a full-time professional carer over a three-year period, and be catered to the living space of each household.
The PDR should:

1. Deliver meals to the involved parties at the requested times at the requested locations.
    User should be able to enter a meal into the PDR.
    User should be able to assign the meal a time and location within the house.
    The PDR should be able to keep meals in a safe environment for up to 4 hours.
2. Collect dishes from the users and store them to be washed.
    User should be able to load cleared dishes into the PDR.
    User should be able to unload cleared dishes from the PDR.
3. Keep track of dietary requirements.
    Users should be able to make “accounts” that represent each person that would be regularly using the PDR.
    User should be able to assign meals entered into the PDR to an existing account.

The PDR's Base functionality is:

1. Navigation through the user's living space.
2. Object sensing and evasion.
3. Environment mapping.
4. Robotic Limb Movement and Fine-Motor Skills.

Additional functionality is:

1. Keep track of dietary requirements.
    - Mark dishes as containing or not-containing specific ingredients or fitting dietary requirements
2. Suggest dishes to make based on ingredients and energy levels
    - User should be able to enter or change a list of ingredients
    - User should be able to enter a meal time
        - Breakfast, Lunch, Dinner, or Snack
    - User should be able to enter an energy level
        - No Energy, Little Energy, Moderate Energy, Full Energy
    - PDR should show a list of recipes that fit all the aforementioned criteria if available
3. PDR should be able to contact emergency services if needed.
4. The PDR should be able to log users activity.
5. User should be able to control the PDR via bluetooth.

Intended Users

For the PDR, "Carers" would be a person with full (over average) physical mobility in all their limbs, that lives with and/or is familiar with their Ward in a way in which they are assured in their wards dietary needs, and can confidently prepare food for them.

While in real life, wards can be anyone, for the purpose of testing the PDR, "Wards" would be a person with limited physical mobility in their arms. They are strong enough to feel themselves but too weak to lift a tray, and have varying levels of physical mobility in their legs.

Research

Supporting Documents

Supporting Software:

In order for the PDR to be user friendly, supporting software needs to be created that would facilitate the use of the PDR - specifically, the creation of representations of users,

environments, and objects would need to be simplified. As such, a basic solution, with a database, will be needed.

Functional Requirements:

Base Functionality:

ID	Category	Requirement	MoSCoW	Comments
FR_BF_01	Navigation through the user's living space.	The PDR must move forward and backward.	Must	
FR_BF_02	Navigation through the user's living space.	The PDR must rotate up to and more than 360 degrees	Must	
FR_BF_03	Navigation through the user's living space.	The PDR must move smoothly over at least two of the following: carpet, wood, or tiled flooring.	Must	
FR_BF_04	Object sensing and evasion.	The PDR must be able to detect objects in front and behind itself	Must	
FR_BF_05	Object sensing and evasion.	The PDR must be able to stop moving when less than 10cm close to a detected object.	Must	
FR_BF_06	Object sensing and evasion.	The PDR should be able to determine an alternate available route to the chosen destination	Should	
FR_BF_07	Object sensing	The PDR could	Could	

ID	Category	Requirement	MoSCoW	Comments
	and evasion.	call for help if no routes are available		
FR_BF_08	Specific Movements and Fine-Motor Skills.	The PDR must be able to open and close the meal storage compartment.	Must	
FR_BF_09	Specific Movements and Fine-Motor Skills.	The PDR could be able to open and close the dish storage compartment	Could	
FR_BF_10	Specific Movements and Fine-Motor Skills.	The PDR could be able to lift and lower the meal base tray	Could	
	Additional Requirements	The PDR must be able to contact emergency services in the event of a medical emergency	Must	

Software Functionality:

ID	Category	Requirement	MoSCoW	Comments
FR_SF_01	Representation of Users	The User must be able to create a new “Person” instance.	Must	
FR_SF_02	Representation of Users	The User must be able to delete an existing	Must	

ID	Category	Requirement	MoSCoW	Comments
		“Person” instance.		
FR_SF_03	Representation of Users	The User must be able to change an existing “Person” instance’s details.	Must	
FR_SF_04	Representation of Meals	The User must be able to create a new “Meal” instance.	Must	
FR_SF_05	Representation of Meals	The User must be able to delete an existing “Meal” instance.	Must	
FR_SF_06	Representation of Meals	The User must be able to assign a time and a room to a new or existing “Meal” instance.	Must	
FR_SF_07	Representation of Meals	The User must be able to assign a dietary restriction to a new or existing “Meal” instance.	Must	
FR_SF_08	Representation of Environment	The User should be able to create a new “Floor” instance.	Should	
FR_SF_09	Representation of Environment	The User should be able to delete an existing “Floor” instance.	Should	

ID	Category	Requirement	MoSCoW	Comments
FR_SF_10	Representation of Environment	The User should be able to change an existing “Floor” instance’s details.	Should	
FR_SF_11	Representation of Environment	The User must be able to create a new “Room” instance.	Must	
FR_SF_12	Representation of Environment	The User must be able to delete an existing “Room” instance.	Must	
FR_SF_13	Representation of Environment	The User must be able to change an existing “Room” instance’s details.	Must	
FR_SF_14	Representation of Environment	The User should be able to assign a “Room” instance to a “Floor” instance	Should	

- Solution for movement + turning and all that good stuff
    - Refining the movement program
- Solution for the more codey-code, more complicated stuff
    -

Non-Functional Requirements

Base Functionality:

ID	Category	Requirement	MoSCoW	Comments
	Speed			
	Reliability			
NFR_BF_01	Useability	The Meal Storage compartment must have a properly installed food-safe insulation fabric on all relevant surfaces.	Must	

Methods

Navigation:

-FR_BF_01

The PDR must move forward and backward.

This requirement is pretty easily met by utilising the VEX V5 drivetrain. The V5 drivetrain has several possible styles depending on the needs of the project. Details on each style can be found at https://kb.vex.com/hc/en-us/articles/360035952771-Creating-a-V5-Drivetrain.

The styles available consist of:

The Standard Drive, which has wheels all in parallel and can be configured to have 2 or 4 motors.

The H-Drive, which has 4 wheels in parallel and an additional fifth running perpendicular.

The Mecanum, which is a 4-wheel drive configuration with the wheels consisting of angled rollers, allowing for easy lateral movement and on-a-point rotation.

The Holonomic, which consists of 4 wheels oriented diagonally at the corners of the drivetrain.

And, The Track Drive, which resembles tank treads, wrapping a track around the wheels of the drivetrain.

Of these options the only one which can be immediately ruled out is the H-Drive, as it would be quite easy for the 5 wheel to become stuck on obstacles around the users home.

-FR_BF_02

The PDR must rotate up to and more than 360 degrees.

So long as the design of the drivechain is not wider than it is long this will be an implicit feature of the design as VEX drivechains are all-wheel drive and allow for asymmetric control over the motors. If the drivechain design does end up being longer than it is wide, then we will likely have to use either a holonomic or tank drive style to ensure that smooth rotation is possible.

-FR_BF_03

The PDR must move smoothly over at least two of the following: carpet, wood, or tiled flooring.

For navigation over multiple surface types an easy out would be to use a tank drive, however, existing products that tackle this problem, such as the roomba, often use an infrared camera on the bottom of the device to detect changes in surface. There is also a very real possibility that the wheels used in the standard drive are already capable of navigating the necessary surfaces around a user's home.

-Conclusion

Of the available options, I believe the Standard Drive would be the best fit for this project. The 4-wheel drive configuration would allow for on-a-point turning in most design cases, and its simplicity would mean that no consideration would have to be given to how it actually drives, unlike the Holonomic and Mecanum styles.

Also, as previously stated, I believe the H-Drive to be unsuitable for this project due to the risks of the fifth wheel becoming caught and halting the PDR.

Specific Movements and Fine-Motor Skills:

Requirements:

ID	Category	Requirement	MoSCoW	Comments
FR_BF_08	Specific Movements and Fine-Motor	The PDR must be able to open and close the	Must	

	Skills.	meal storage compartment.		
FR_BF_09	Specific Movements and Fine-Motor Skills.	The PDR could be able to open and close the dish storage compartment	Could	
FR_BF_10	Specific Movements and Fine-Motor Skills.	The PDR could be able to lift and lower the meal base tray	Could	

FR_BF_08 & FR_BF_09:

Overview:

"The PDR must be able to open and close the meal storage compartment."

- Our design requires that the PDR have several storage compartments in order to store food, drinks, and other items. The main ("meal") storage compartment should be the biggest of all compartments, and should be sealed off from outside the PDR when not in use.
- The PDR must be able to open and close the meal storage compartment in the following situations:
    - When the PDR has reached the specified destination
        - Delivering food
        - Returning used dishes
    - At the user's discretion

This requirement is mostly facilitated by the design of the robot rather than any software-dependant solutions, however, there is the matter of keeping the container food-safe while also ensuring ease of movement.
Therefore, the main contention points of this requirement are:

- How can I keep the food carried in this compartment within +/- 5°C of the temperature it had when entering the compartment for at least 2 hours?
    - Lining the compartment with food-safe material?
- How can I ensure food safety without reducing functionality?

How can I keep the food carried in this compartment within +/- 5°C of the temperature it had when entering the compartment for at least 2 hours?:

Temperature Controlled Packaging

Lining the casing with food-safe material could be a viable solution to precisely controlling the temperature of the food.

Food-Safe Material	**Positives**	**Negatives**	**Conclusion**	**Notes**
Insulated box liners	Lightweight Eco-friendly	May not be applicable to repeated use Cannot be washed easily	Would not be suitable for the final product due to its inability to be cleaned, but depending on the ease of procurement it may be useful for prototyping	
PLA Insulation	Comes in various forms Reuseable Washable	Would be difficult to install Can be bulky	May be suitable for the final product if a way to reliably install it can be found.	
Aluminum Foil Lining	Impermeable to oxygen and moisture Easy to install Easy to procure	Chemical corrosion is possible on contact with acidic or alkaline substances	Aluminium foil lining would be both the easiest to procure and the fastest to install, and as such is suitable for both prototyping and the final product.	

How can I ensure food safety without reducing functionality?:

When making the case for the robot, using food-safe plastic or lining the casing with food-safe material could both be reasonable solutions to this issue.

Using Food-Safe 3D Printing Filament

The general consensus is that no 3D printed material is completely food safe, due to the bacteria colonies that can grow in between the layers of the print.

Filament	**Positives**	**Negatives**	**Conclusion**	**Notes**
PolyLactic Acid (PLA)	Affordable Widely Available Many colours Easy to use Adaptable Nice surface finish	Limited Heat Resistance Brittle	PLA would be most useful when prototyping, but not when creating the final project.	Pure PLA is considered food safe, but the inclusion of colorants or additives in many commercially available roles make it unsafe for use.
Acrylonitrile Butadiene Styrene (ABS)	Impact resistance Durability Versatile Strong	Limited Heat Resistance Flammable Poor Weather Resistance Dimensional changes overtime due to temperature warping Less available than PLA	ABS is also a valid choice for prototyping, however, it is a worse choice for a final product than PLA as it is inherently not food-safe.	

Conclusion:

Having the case of the PDR be 3D-printed with PLA plastic and lined with aluminium foil lining would be more beneficial to development.
However, during the prototyping stages, removing the lining and using pure PLA would be more cost-effective and less time consuming.

FR_BF_10:

Overview:

"**The PDR could be able to lift and lower the meal base tray.**"

- The design of the PDR highlights the need for the meal compartment in the PDR to include a tray where food can be set. While this could just be the bottom of the compartment, if the tray was able to be lifted and lowered by the machine, it would take the onus off the ward, who may not have the upper body strength or dexterity needed to take the food out themselves.
- The main concern in this case is determining what payload capacity is needed for our robot and how we can ensure reliability. While it is unlikely that the PDR would need anything substantial, the edge cases are particularly important to test, as there is a substantial risk of injury if something goes wrong in the lifting and lowering process.
- Additionally, the secondary concern of 'the consistent orientation of the tray while moving' may also need to be addressed.

Consistent Tray Orientation:

While the orientation of the tray may sound like an insignificant issue for the PDR, it is extremely important, as it ensures that nothing is broken nor damaged during lifting and lowering. While the orientation of the tray doesn't change through movement and therefore there is no point of contention, there is a possibility that the level of the tray may become unbalanced.

There are many structural ways to ensure leveling, but the most available and efficient method would be to add a gear mechanism that would lift the tray up from at least one point on each side. Alternatively, a pulley system could also be used, though it would provide no clear advantages to using only a gear mechanism.